

Time Series Analyses for Social Data

Conceptual Questions

Please write a three to ten sentence explanation for **ONE** of the two questions below.

1. **Time leakage and honest evaluation.** In the coding tutorial, you computed RMSE under (i) a random train/test split and (ii) a time-based split (train on past, test on future).
 - Explain what *time leakage* is and why random splits usually inflate model performance for time-indexed data.
 - Explain why “train on past, test on future” is the default evaluation rule.
 - Describe what *rolling-origin* (backtesting) adds beyond a single time split.
2. **Causal timing designs and placebos.** Interrupted Time Series (ITS) is often used when an intervention occurs at a known time t_0 but randomization is not available.
 - State one key identification assumption behind ITS.
 - Explain the difference between a “level change” and a “trend change” at t_0 .
 - Propose one placebo diagnostic and explain what failure would look like (e.g., a fake intervention date, pre-trend check).

Applied Exercises

Use the code in `timeseries.R` and the lecture slides to answer the following questions.

bigskip

3. **Decomposition: trend + seasonality + residual (save a figure).** Extend the synthetic daily time series in `timeseries.R` by decomposing it into components.
 - Convert the daily series to a `ts` object with weekly frequency (`frequency = 7`).
 - Run a seasonal-trend decomposition (e.g., `stl()`).
 - Create a single figure showing the observed series and the decomposed components.
 - Save the figure as `outputs/figures/decomposition.png`.
4. **Rolling-origin evaluation (backtesting) for forecasting RMSE.** The lecture emphasizes that evaluation should mimic deployment (fit on past, predict future). Implement a rolling-origin backtest that produces *many* out-of-sample errors instead of a single split.
 - Choose a forecast horizon $h = 1$ day ahead.
 - Choose an initial training window (e.g., first 300 days).
 - For each time t from the end of the initial window to the end of the series:
 - (a) fit an AR(1) model to data up to t (use `arima(..., order=c(1,0,0))` or `forecast::auto.arima` if available),
 - (b) forecast y_{t+1} ,
 - (c) store the one-step-ahead error $e_{t+1} = y_{t+1} - \hat{y}_{t+1}$.
 - Compute RMSE across the backtest errors and compare it to the single time-split RMSE from the tutorial.
 - Save:

- a CSV of the backtest errors (`date`, `y`, `yhat`, `error`) as `outputs/tables/backtest_errors.csv`, and
 - a line plot of y_t and $\hat{y}_t$ over the test region as `outputs/figures/backtest_forecast.png`.
5. **Interrupted Time Series (ITS): level and slope change + placebo date.** Create a synthetic intervention at time t_0 on top of a trend (use the lecture’s ITS framing).
- Pick an intervention date t_0 around the middle of the series (report your choice).
 - Create an intervention indicator $I[t \geq t_0]$ and a post-intervention time counter $(t - t_0) I[t \geq t_0]$.
 - Fit an ITS regression:

$$y_t = \alpha + \delta t + \tau_1 I[t \geq t_0] + \tau_2 (t - t_0) I[t \geq t_0] + \varepsilon_t$$

- Plot three lines on the same figure:
 - observed y_t ,
 - fitted values from the ITS model,
 - counterfactual values setting $\tau_1 = \tau_2 = 0$ (pre-period trend extended forward).
- Run a placebo ITS with a *fake* intervention date in the pre-period and report whether it produces a large “effect.” (Briefly interpret.)
- Save:
 - coefficient table for the real ITS and placebo ITS as `outputs/tables/its_results.csv`, and
 - your main ITS figure as `outputs/figures/its_plot.png`.

Challenge Question (Optional — if you finish early)

Choose **ONE** option.

- (a) **Switchback experiment (time-block randomization).** Simulate a switchback design where treatment alternates by time blocks (e.g., every 7 days) and estimate the effect.
- Create a block assignment over time (alternating 0/1).
 - Generate an outcome with trend + noise + a treatment effect that operates only during treated blocks.
 - Estimate the difference in means across treated vs control blocks.
 - Placebo: shift assignment by 1 block (lag the assignment) and show the estimated effect shrinks toward 0.
 - Save a plot of the time series with shaded treated blocks as `outputs/figures/switchback.png`.
- (b) **Sequential testing / peeking inflation (simulation).** Simulate repeated monitoring where you compute a p-value every day after a minimum start date.
- Under a true null (no treatment effect), compute daily “p-values” for a naive treated vs control comparison.
 - Show that the probability of getting at least one p-value < 0.05 is much larger than 0.05.
 - Briefly explain why this happens and name one control strategy (alpha spending, pre-specified stopping).